

Ex.No:1b**Date:18/8/25**

Basic Networking Commands in Windows operating system

Aim:

To study and execute basic networking commands in Windows operating system.

1. cd

SYNOPSIS: CD [/D] [drive:][path]

CD [..]

DESCRIPTION: Displays the name of or changes the current directory.

OUTPUT:

```
C:\Users\ponsu>cd Desktop  
C:\Users\ponsu\Desktop>
```

2. cls

SYNOPSIS: cls

DESCRIPTION: Clears the screen.

OUTPUT:

```
C:\Users\ponsu\Desktop>
```

3. ipconfig

SYNOPSIS: ipconfig [/allcompartments]

DESCRIPTION: The default is to display only the IP address, subnet mask and default gateway for each adapter bound to TCP/IP.

OUTPUT:

```
C:\Users\ponsu\Desktop>ipconfig

Windows IP Configuration

Unknown adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Unknown adapter Norton VPN:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::be2b:3d6f:e027:c8%18
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Unknown adapter OpenVPN Connect DCO Adapter:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2401:4900:1cd4:41d7:96d6:f7e0:7ac0:1d2f
    Temporary IPv6 Address. . . . . : 2401:4900:1cd4:41d7:2de0:6489:130b:3d12
    Link-local IPv6 Address . . . . . : fe80::25b1:174a:723b:da2e%21
    IPv4 Address. . . . . : 192.168.1.3
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1%21
                                192.168.1.1

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::3dc3:21be:bfef:1b36%56
    IPv4 Address. . . . . : 172.24.224.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . :
```

4. echo

SYNOPSIS: ECHO [ON | OFF]

ECHO [message]

DESCRIPTION: Displays messages, or turns command-echoing on or off.

OUTPUT:

```
C:\Users\ponsu\Desktop>echo "hi"  
"hi"
```

5. del

SYNOPSIS: DEL [/P] [/F] [/S] [/Q] [/A[:attributes]] names

ERASE [/P] [/F] [/S] [/Q] [/A[:attributes]] names

DESCRIPTION: Deletes one or more files.

OUTPUT:

```
C:\Users\ponsu\Desktop>del test.txt
```

6. find

SYNOPSIS: FIND [/V] [/C] [/N] [/I] [/OFF[LINE]] "string"
[[drive:][path]filename[...]]

DESCRIPTION: Searches for a text string in a file or files.

OUTPUT:

```
C:\Users\ponsu\Desktop>find "Hello" tt.txt  
  
----- TT.TXT  
Hello There!
```

7. tasklist

SYNOPSIS: TASKLIST [/S system [/U username [/P [password]]]]

[/M [module] | /SVC | /V] [/FI filter] [/FO format] [/NH]

DESCRIPTION: This tool displays a list of currently running processes on either a local or remote machine.

OUTPUT:

```
C:\Users\ponsu\Desktop>tasklist

Image Name                      PID Session Name        Session#    Mem Usage
=====
System Idle Process             0 Services             0             8 K
System                           4 Services             0            7,844 K
Secure System                   188 Services           0           62,040 K
Registry                        232 Services           0           29,436 K
smss.exe                        792 Services           0             792 K
csrss.exe                      1084 Services           0            4,452 K
wininit.exe                     1172 Services           0            4,316 K
services.exe                   1320 Services           0           11,152 K
lsass.exe                      1340 Services           0            2,072 K
lsass.exe                      1348 Services           0           26,476 K
svchost.exe                     1476 Services           0           34,048 K
fontdrvhost.exe                1496 Services           0            1,800 K
WUDFHost.exe                   1608 Services           0           10,112 K
```

8. systeminfo

SYNOPSIS: SYSTEMINFO [/S system [/U username [/P [password]]]]

[/FO format] [/NH]

DESCRIPTION: This tool displays operating system configuration information for a local or remote machine, including service pack levels.

OUTPUT:

```
C:\Users\ponsu\Desktop>systeminfo | findstr "Time Zone"
System Boot Time:          01-10-2025, 13:13:07
Time Zone:                 (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
```

9. tracert

SYNOPSIS: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]

[-R] [-S srcaddr] [-4] [-6] target_name

DESCRIPTION: Traces the route packets take to a destination, showing each hop and its response time to diagnose network issues.

OUTPUT:

```
C:\Users\ponsu\Desktop>tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  1     6 ms     3 ms     3 ms   RTK_GW [192.168.1.1]
  2    25 ms     6 ms     6 ms   abts-tn-dynamic-1.132.78.171.airtelbroadband.in [171.78.132.1]
  3    14 ms    16 ms     8 ms   125.17.36.41
  4    19 ms     9 ms     8 ms   182.79.141.35
  5     6 ms     6 ms     8 ms   182.79.191.222
  6   393 ms    10 ms    11 ms   116.119.161.147
  7    14 ms    10 ms     8 ms   72.14.197.10
  8    15 ms     8 ms     7 ms   216.239.51.91
  9    14 ms    11 ms     8 ms   142.251.55.41
 10    16 ms    10 ms    10 ms   dns.google [8.8.8.8]

Trace complete.
```

10. ping

SYNOPSIS: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
[-r count] [-s count] [[-j host-list] | [-k host-list]]
[-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
[-4] [-6] target_name

DESCRIPTION: Ping tests network connectivity and measures response time between two devices.

OUTPUT:

```
C:\Users\ponsu\Desktop>ping www.google.com

Pinging www.google.com [2404:6800:4002:815::2004] with 32 bytes of data:
Reply from 2404:6800:4002:815::2004: time=159ms
Reply from 2404:6800:4002:815::2004: time=164ms
Reply from 2404:6800:4002:815::2004: time=162ms
Reply from 2404:6800:4002:815::2004: time=75ms

Ping statistics for 2404:6800:4002:815::2004:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 75ms, Maximum = 164ms, Average = 140ms
```

Result:

Thus the above windows commands were executed.